Lecture 16: Group Velocity, Probability Current, and Potential Steps

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Lecture 15 showed that a free Gaussian packet moves while spreading. We now connect the motion of its center to the dispersion relation, derive the probability current, and use current to distinguish reflection from transmission at a potential step.

Questions.

- What do phase velocity and group velocity describe?
- How does probability flow in one dimension?
- Why can a wave function penetrate into a forbidden region without being transmitted?
- Why can a particle with $E > V_0$ still be reflected by a potential step?

Lecture roadmap. We first review the moving Gaussian packet and identify its group velocity. We then derive the continuity equation and probability current. Finally, we match stationary wave functions at a potential step for $0 < E < V_0$ and $E > V_0$, and interpret the results for a wave packet.

Review from Lecture 15

For a freely moving Gaussian packet, the probability density is

$$|\psi(x,t)|^2 = \frac{1}{\sqrt{\pi}\,a(t)} \exp\left[-\frac{(x-x_c(t))^2}{a^2(t)}\right], \tag{1}$$

with

$$a(t) = a\sqrt{1+(t/\tau)^2}, \qquad \tau = \frac{ma^2}{\hbar}. \tag{2}$$

Its center is

$$\boxed{x_c(t) = \langle x\rangle_t = x_0 + \frac{p_0}{m}t = x_0 + \frac{\hbar k_0}{m}t.} \tag{3}$$

The packet center translates while the packet spreads. We now relate the translation speed to the free-particle dispersion relation.

1 Group velocity and phase velocity

For a free particle,

$$E = \frac{\hbar^2 k^2}{2m}, \qquad \omega(k) = \frac{E}{\hbar} = \frac{\hbar k^2}{2m}. \tag{4}$$

A plane-wave phase crest satisfies $kx - \omega t =$ constant. For $k_0 \neq 0$, its speed, the *phase velocity*, is therefore

$$v_p = \frac{\omega(k_0)}{k_0} = \frac{\hbar k_0}{2m} = \frac{p_0}{2m}. \tag{5}$$

The envelope of a wave packet is built from nearby wave numbers. Its speed, the *group velocity*, is

$$v_g = \left.\frac{\mathrm{d}\omega}{\mathrm{d}k}\right|_{k_0} = \frac{\hbar k_0}{m} = \frac{p_0}{m}. \tag{6}$$

Thus, for a free nonrelativistic particle,

$$\boxed{v_g = 2v_p.} \tag{7}$$

The group velocity agrees with the velocity of the packet center:

$$\boxed{x_c(t) = x_0 + v_g t = x_0 + \frac{p_0}{m}t.} \tag{8}$$

This is the classical free-particle trajectory for the center of the packet. Quantum mechanics also predicts that the packet spreads, so its full spatial distribution is not a classical point trajectory.

	Phase velocity	**Group velocity**
describes	phase crests of the carrier	motion of the packet envelope
definition	$v_p = \omega/k$	$v_g = \mathrm{d}\omega/\mathrm{d}k$
free particle	$\hbar k_0/(2m)$	$\hbar k_0/m$

1.1 What the moving-packet simulation shows

Plotting both $\mathrm{Re}\,\psi$ and $|\psi|$ distinguishes the carrier oscillations from the packet envelope. The phase velocity describes the carrier plane-wave phase, while the group velocity describes the motion of the envelope and packet center. A visible crest is not a separate plane-wave component.

The simulation uses $\sigma = \Delta x(0) = a/\sqrt{2}$. For the moving Gaussian,

$$\Delta p = \frac{\hbar}{2\sigma}, \qquad E_0 = \frac{\hbar^2 k_0^2}{2m}, \qquad \langle H\rangle = E_0 + \frac{\hbar^2}{8m\sigma^2}. \tag{9}$$

The main observations are:

- As time increases, the center moves at v_g while the width increases. The peak falls, but the total probability remains one.
- Reducing the initial width increases the momentum and energy spreads and causes faster spreading. Even at fixed E_0, it increases the mean energy.
- A plane wave has uniform magnitude and one definite momentum. It has no localized envelope or packet center.

2 Probability density and probability current

For a real potential $V(x)$, the time-dependent Schrödinger equation and its complex conjugate are

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V\psi, \tag{10}$$

$$-i\hbar\frac{\partial\psi^*}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi^*}{\partial x^2} + V\psi^*. \tag{11}$$

Multiply the first equation by ψ^*, the second by ψ, and subtract. The potential terms cancel because V is real:

$$\frac{\partial|\psi|^2}{\partial t} = -\frac{\hbar}{2mi}\frac{\partial}{\partial x}\left(\psi^*\frac{\partial\psi}{\partial x} - \psi\frac{\partial\psi^*}{\partial x}\right). \tag{12}$$

Define the probability density and probability current by

$$\boxed{\rho(x,t) = |\psi(x,t)|^2, \qquad j(x,t) = \frac{\hbar}{2mi}\left(\psi^*\frac{\partial\psi}{\partial x} - \psi\frac{\partial\psi^*}{\partial x}\right).} \tag{13}$$

Then

$$\boxed{\frac{\partial\rho}{\partial t} + \frac{\partial j}{\partial x} = 0.} \tag{14}$$

This is the one-dimensional continuity equation. For any interval $[x_L, x_R]$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_{x_L}^{x_R}\rho\,\mathrm{d}x = j(x_L,t) - j(x_R,t). \tag{15}$$

The probability inside the interval changes only through probability flowing across its boundaries.

2.1 Current for traveling waves

For a stationary state $\psi(x,t) = \phi(x)e^{-iEt/\hbar}$, the time-dependent phase cancels from the current. We may therefore calculate it directly from $\phi(x)$:

$$j[\phi] = \frac{\hbar}{2mi}\left(\phi^*\phi' - \phi\phi'^*\right). \tag{16}$$

For a right-moving plane wave $\phi = Ae^{ikx}$, $\phi' = ik\phi$, giving

$$j_{\text{right}} = \frac{\hbar}{2mi}(ik|\phi|^2 + ik|\phi|^2) = \frac{\hbar k}{m}|A|^2. \tag{17}$$

For a left-moving plane wave $\phi = Be^{-ikx}$,

$$j_{\text{left}} = -\frac{\hbar k}{m}|B|^2. \tag{18}$$

The sign indicates the direction of probability flow. In particular, a right-moving wave carries positive current and a left-moving wave carries negative current.

By contrast, a decaying exponential $\phi = De^{-\kappa x}$ carries no current. Indeed, $\phi' = -\kappa\phi$, so

$$j = \frac{\hbar}{2mi}\left[-\kappa|\phi|^2 - (-\kappa|\phi|^2)\right] = 0. \tag{19}$$

The wave function and its probability density can be nonzero even when there is no probability current.

Questions.

Question 1. Can a nonzero decaying exponential $\phi(x) = De^{-\kappa x}$, with real $\kappa > 0$, carry probability current?

Answer and explanation. No. Its derivative is a real multiple of the wave function, $\phi' = -\kappa\phi$. Therefore the two terms $\phi^*\phi'$ and $\phi\phi'^*$ in the current are equal and cancel:

$$j = \frac{\hbar}{2mi}\left(\phi^*\phi' - \phi\phi'^*\right) = 0.$$

The exponential tail can have nonzero probability density, but it does not transport probability to infinity.

2.2 Reflection and transmission

For incidence from the left, let $j_{\text{in}} > 0$ be the incident current, $j_{\text{refl}} < 0$ the reflected current, and $j_{\text{trans}} > 0$ the transmitted current. Define the reflection and transmission probabilities as

$$\boxed{R = \frac{|j_{\text{refl}}|}{j_{\text{in}}}, \qquad T = \frac{j_{\text{trans}}}{j_{\text{in}}}.} \tag{20}$$

These are ratios of probability currents, not generally just ratios of squared wave amplitudes. Current conservation gives $R + T = 1$ when there are no other outgoing channels.

3 Potential step with $0 < E < V_0$

Consider a step that extends to the right without end:

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & x > 0, \end{cases} \qquad 0 < E < V_0. \tag{21}$$

Semi-infinite describes the extent of the region $x > 0$; it does not mean that V_0 is infinite.

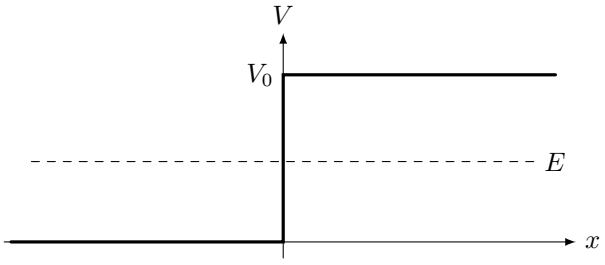


Figure 1: A semi-infinite upward potential step with $0 < E < V_0$.

Define the positive wave number and decay constant

$$k = \frac{\sqrt{2mE}}{\hbar}, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}. \tag{22}$$

The stationary Schrödinger equation has different forms in the two regions:

region	equation	general solution
$x < 0$	$\phi'' + k^2\phi = 0$	$\phi_I = Ae^{ikx} + Be^{-ikx}$
$x > 0$	$\phi'' - \kappa^2\phi = 0$	$\phi_{II} = Ce^{\kappa x} + De^{-\kappa x}$

(23)

For incidence from the left, Ae^{ikx} is the incident wave and Be^{-ikx} is the reflected wave. Both are oscillatory and remain bounded as $x \to -\infty$, so both must be kept. In region II, $e^{\kappa x}$ grows without bound as $x \to +\infty$; its coefficient must vanish. These extended stationary scattering states are the building blocks for normalizable incident packets. The allowed scattering solution is therefore

$$\phi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0, \\ De^{-\kappa x}, & x > 0. \end{cases} \tag{24}$$

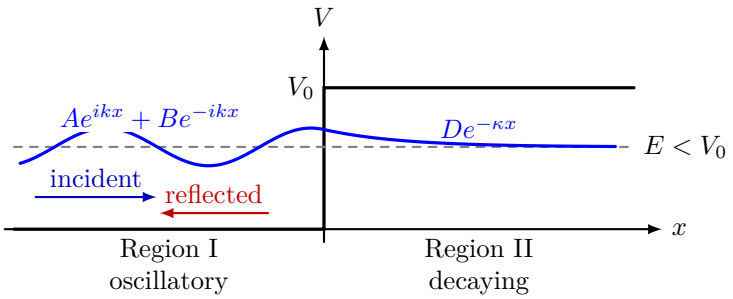


Figure 2: Schematic real part of a stationary scattering state for $E < V_0$. The oscillation and decaying tail are drawn about the energy line; the sketch is not a probability density.

3.1 Matching and reflection

At a finite step, both the wave function and its first derivative are continuous at $x = 0$. Hence

$$A + B = D, \qquad ik(A - B) = -\kappa D. \tag{25}$$

Define the reflection amplitude r and the right-region amplitude ratio t:

$$r \equiv \frac{B}{A}, \qquad t \equiv \frac{D}{A}. \tag{26}$$

Dividing the matching equations by A gives

$$1 + r = t, \qquad ik(1 - r) = -\kappa t. \tag{27}$$

Substitute $t = 1 + r$ into the second equation:

$$ik(1 - r) = -\kappa(1 + r), \tag{28}$$

$$ik + \kappa = r(ik - \kappa), \tag{29}$$

$$r = \frac{ik + \kappa}{ik - \kappa} = \frac{k - i\kappa}{k + i\kappa}. \tag{30}$$

Then

$$\boxed{r = \frac{k - i\kappa}{k + i\kappa}, \qquad t = 1 + r = \frac{2k}{k + i\kappa}.} \tag{31}$$

Here t is the amplitude ratio of the decaying tail. It is not a transmission probability.

The incident and reflected currents are

$$j_{\text{in}} = \frac{\hbar k}{m}|A|^2, \qquad j_{\text{refl}} = -\frac{\hbar k}{m}|B|^2. \tag{32}$$

Therefore

$$R = \frac{|j_{\text{refl}}|}{j_{\text{in}}} = \left|\frac{B}{A}\right|^2 = |r|^2 \tag{33}$$

$$= \frac{|k - i\kappa|^2}{|k + i\kappa|^2} = \frac{k^2 + \kappa^2}{k^2 + \kappa^2} = 1. \tag{34}$$

The current to the right of the step vanishes:

$$\phi'_{II} = -\kappa\phi_{II} \quad \Longrightarrow \quad j_{II} = \frac{\hbar}{2mi}\left(-\kappa|\phi_{II}|^2 + \kappa|\phi_{II}|^2\right) = 0. \tag{35}$$

Thus

$$\boxed{R = 1, \qquad T = 0.} \tag{36}$$

The wave function penetrates into $x > 0$, but no probability current flows toward $+\infty$. The amplitude decays over the penetration length

$$\boxed{\ell = \frac{1}{\kappa} = \frac{\hbar}{\sqrt{2m(V_0 - E)}}.} \tag{37}$$

This is the distance over which the wave-function amplitude falls by $1/e$; the probability density decays as $e^{-2\kappa x}$. As $V_0 \to \infty$ at fixed E, $\kappa \to \infty$, so the penetration length tends to zero and $r \to -1$.

Questions.

Question 2. The wave function is nonzero for $x > 0$. Has probability been transmitted to $+\infty$?

Answer and explanation. No. The right-hand solution is a decaying exponential, not an outgoing traveling wave. Its current is zero, so $T = 0$, even though its probability density is nonzero near the interface.

3.2 Physical picture from a wave packet

A localized packet with energies below V_0 approaches the step and develops a decaying tail in $x > 0$. At late times there is no separate outgoing packet on the right; the packet is reflected back to the left. This is consistent with $R = 1$ and $T = 0$. A packet has a range of energies, so this conclusion applies to a packet whose energy distribution lies below V_0; components above V_0, if present, can be transmitted.

4 Potential step with $E > V_0$

For the same step, let the energy exceed the step height:

$$E > V_0, \qquad k_1 = \frac{\sqrt{2mE}}{\hbar}, \qquad k_2 = \frac{\sqrt{2m(E - V_0)}}{\hbar}. \tag{38}$$

Both regions support oscillatory solutions:

$$\phi_I = Ae^{ik_1 x} + Be^{-ik_1 x}, \qquad \phi_{II} = Ce^{ik_2 x} + De^{-ik_2 x}. \tag{39}$$

For incidence only from the left, there is no incoming wave from $+\infty$. We therefore set $D = 0$, obtaining

$$\phi(x) = \begin{cases} Ae^{ik_1 x} + Be^{-ik_1 x}, & x < 0, \\ Ce^{ik_2 x}, & x > 0. \end{cases} \tag{40}$$

The term $Ae^{ik_1 x}$ is incident, $Be^{-ik_1 x}$ is reflected, and $Ce^{ik_2 x}$ is transmitted.

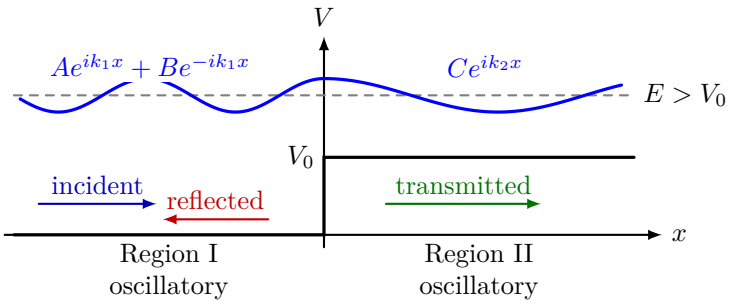


Figure 3: Schematic stationary wave for $E > V_0$. The transmitted wave has a longer wavelength because $k_2 < k_1$.

4.1 Matching and amplitudes

Continuity of ϕ and ϕ' at $x = 0$ gives

$$A + B = C, \qquad k_1(A - B) = k_2 C. \tag{41}$$

Define the reflection and transmission amplitudes

$$r \equiv \frac{B}{A}, \qquad t \equiv \frac{C}{A}. \tag{42}$$

Then

$$1 + r = t, \qquad k_1(1 - r) = k_2 t. \tag{43}$$

Substituting $t = 1 + r$ gives

$$k_1(1 - r) = k_2(1 + r), \tag{44}$$

$$k_1 - k_2 = r(k_1 + k_2), \tag{45}$$

and hence

$$\boxed{r = \frac{k_1 - k_2}{k_1 + k_2}, \qquad t = 1 + r = \frac{2k_1}{k_1 + k_2}.} \tag{46}$$

4.2 Current probabilities

The incident, reflected, and transmitted currents are

$$j_{\text{in}} = \frac{\hbar k_1}{m}|A|^2, \qquad j_{\text{refl}} = -\frac{\hbar k_1}{m}|B|^2, \qquad j_{\text{trans}} = \frac{\hbar k_2}{m}|C|^2. \tag{47}$$

Thus the reflection probability is

$$\boxed{R = \frac{|j_{\text{refl}}|}{j_{\text{in}}} = |r|^2 = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2.} \tag{48}$$

For transmission, the different wave numbers give different current per unit amplitude:

$$\boxed{T = \frac{j_{\text{trans}}}{j_{\text{in}}} = \frac{k_2}{k_1}|t|^2 = \frac{4k_1k_2}{(k_1 + k_2)^2}.} \tag{49}$$

The factor k_2/k_1 is essential; in general, $T \neq |t|^2$. The two probabilities add to one:

$$R + T = \frac{(k_1 - k_2)^2 + 4k_1k_2}{(k_1 + k_2)^2} = 1. \tag{50}$$

Since $k_2/k_1 = \sqrt{1 - V_0/E}$, these can also be written as

$$R = \left(\frac{1 - \sqrt{1 - V_0/E}}{1 + \sqrt{1 - V_0/E}}\right)^2, \qquad T = \frac{4\sqrt{1 - V_0/E}}{(1 + \sqrt{1 - V_0/E})^2}. \tag{51}$$

4.3 Why reflection vanishes at high energy

When $E \gg V_0$, define $\delta = V_0/E \ll 1$. Then

$$\frac{k_2}{k_1} = \sqrt{1 - \delta} \approx 1 - \frac{\delta}{2}. \tag{52}$$

Consequently,

$$r = \frac{1 - \sqrt{1 - \delta}}{1 + \sqrt{1 - \delta}} \approx \frac{\delta}{4} = \frac{V_0}{4E}. \tag{53}$$

Therefore

$$\boxed{R \approx \left(\frac{V_0}{4E}\right)^2 \longrightarrow 0, \qquad T = 1 - R \longrightarrow 1.} \tag{54}$$

At high energy the step is small compared with the particle's energy. The wave numbers on its two sides become nearly equal, so the wavelength mismatch and the reflected amplitude both become small. The step remains present; it is its relative effect that vanishes as E/V_0 grows.

Questions.

Question 3. If $E > V_0$, why is R generally nonzero?

Answer and explanation. The wave number changes at the abrupt step: $k_1 \neq k_2$. A single wave traveling to the right cannot satisfy continuity of both ϕ and ϕ' at the interface, so a reflected component is needed. This above-step reflection is a quantum wave-matching effect; classically, a particle with $E > V_0$ would cross the step.

4.4 Physical picture from a wave packet

An incident packet splits into a reflected packet and a transmitted packet. Because $k_2 < k_1$, the transmitted packet has a longer wavelength. In a constant-potential region the dispersion relation is $\omega = V/\hbar + \hbar k^2/(2m)$, so its group velocity is

$$v_{g,1} = \frac{\hbar k_1}{m}, \qquad v_{g,2} = \frac{\hbar k_2}{m}. \tag{55}$$

Thus the transmitted packet moves more slowly on the higher-potential side. Reflection is a quantum effect, and it becomes weaker as the energy becomes large compared with V_0.

For a downward step, the transmitted packet instead gains kinetic energy: its wavelength becomes shorter and its group speed increases. A reflected packet can still appear because the wave number changes at the interface.

Lecture 16 summary

- For a free particle, the packet center travels at the group velocity:

$$v_g = \frac{\mathrm{d}\omega}{\mathrm{d}k} = \frac{p_0}{m} = 2v_p, \qquad x_c(t) = x_0 + v_g t.$$

- Probability is locally conserved:

$$\frac{\partial \rho}{\partial t} + \frac{\partial j}{\partial x} = 0, \qquad j = \frac{\hbar}{2mi}\left(\psi^* \partial_x \psi - \psi \partial_x \psi^*\right).$$

- For a step with $0 < E < V_0$, the wave function has a decaying tail in the right region, but the tail carries no current:

$$R = 1, \qquad T = 0.$$

- For a step with $E > V_0$, both reflection and transmission occur:

$$R = \left(\frac{k_1 - k_2}{k_1 + k_2}\right)^2, \qquad T = \frac{4k_1 k_2}{(k_1 + k_2)^2}, \qquad R + T = 1.$$

- For $E \gg V_0$, the wave numbers nearly match and $T \to 1$.

Suggested reading. Griffiths and Schroeter, *Introduction to Quantum Mechanics*, 3rd ed., the sections on probability current and one-dimensional scattering from a potential step. Review the distinction between wave amplitudes and probability-current ratios.